Ashutosh Mishra

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Research Appointments

Istituto Italiano di Tecnologia (IIT) **Aug. 2026 – Present**
Postdoctoral Researcher [Dynamic Legged Systems Lab] - Supervisor: Dr. Claudio Semini *Genova, Italy*

- Developing learning-based locomotion controllers for quadrupedal robots applied to space exploration, under the ALTEC project.
- Building GPU-accelerated RL training and sim-to-real transfer pipelines on Isaac Sim, Isaac Lab, and NVIDIA Newton for deployment on legged hardware.

Education

Tohoku University **Oct. 2023 – Mar 2026**
Doctoral Researcher [Space Robotics Lab] - Govt. MEXT Scholar *Sendai, Japan*

Pune University **Aug. 2021 – July 2023**
Master of Science in Electronics - 9.6/10 CGPA *Pune, India*

University of Delhi **Aug. 2018 – July 2021**
Bachelor of Science in Electronics - 8.6/10 CGPA *Delhi, India*

Experiences

International Space University, Space Studies Program 2024 **Jun 2024 – Aug 2024**
Project Management Team Lead and Lead Editor *NASA JSC and Rice University, Houston, USA*

- Led the Project Management Team and served as Lead Editor for the International Lunar University project.
- Worked under the supervision of Dr. Jacob Cohen, Chief Scientist at NASA Ames.
- Collaborated with international teams to propose a strategic vision for establishing a Lunar University.

Wobot Intelligence INC. **Jan 2023 – June 2023**
Computer Vision Intern *California, USA*

- Developed and deployed various neural network models optimized for edge computing.
- Applied Computer Vision techniques for video intelligence and analytics using Nvidia Jetson.
- Worked with Python, Nvidia Triton Server, Docker, OpenCV, TensorFlow, and Keras.

EVI Technologies Pvt. Ltd. **Jan 2021 – Dec 2021**
Cloud Platform Engineer *New Delhi, India*

- Built and deployed a central cloud platform system for controlling and managing electric vehicle charging stations.
- Managed Linux-based servers and backend infrastructure for EVIT and the Centre for Automotive Research and Tribology (CART) at IIT Delhi.
- Worked with Python, Django, Apache, AWS, and MySQL.

JNE Smart Technologies Pvt. Ltd. **May 2020 – Dec 2020**
Embedded Engineer Intern *Ghaziabad, India*

- Developed and deployed a health kiosk machine to monitor government employees' health status in Meghalaya, India.
- Worked on AgriTech control systems for autonomous agricultural field monitoring and irrigation.
- Used Raspberry Pi, Python, Flask, and embedded system design tools.

VHCPL Automation India Pvt. Ltd. **July 2019 – Sept 2019**
Embedded Engineer Intern *Delhi, India*

- Developed and deployed IoT devices for automation tasks.
- Worked on real-life problem-solving using embedded systems, Raspberry Pi, Esp32Cam, and ARM controllers.

Projects

Self-Evolving AI Robot System for Lunar Exploration **Oct 2023 – Present**

- Developing autonomous, self-evolving AI robot systems for lunar exploration and outpost construction under Prof. Kazuya Yoshida at the Space Robotics Lab, Tohoku University, as part of the JST's Moonshot Goal 3 project.

Moonbot APEX: Agentic Planning and Execution Framework **Jun 2025 – Jul 2025**

JAXA Space Exploration Test Field, Sagamihara

- Conducted full-scale field validation of the Moonbot APEX framework for modular lunar robots under Moonshot Goal 3, testing agentic planning and execution capabilities in lunar-like terrain and lighting conditions.

Modular RL Policy Implementation on Reconfigurable Robots **Feb 2025 – Mar 2025**

JAXA Space Exploration Test Field, Sagamihara

- Tested reinforcement learning-based control policies on morphology-varying modular robots, validating adaptive control and robustness across multiple reconfigurable morphologies.

Plug and Play Modular Sensor Box **Sept 2025 – Oct 2025**

Abukuma Cave Field Testing

- Designed and Developed Modular Plug and Play Sensor Box for MoonBot Platform, with sensor fusion of Livox Mid 360 LiDAR, two fish eye cameras and Intel Realsense D435i camera. Field tested with mapping Abukuma Cave for lunar cave simulation work.

Enhanced Manipulator Control for Solar Infrastructure Deployment **Oct 2024 – Nov 2024**

JAXA Space Exploration Test Field, Sagamihara

- Performed human-in-the-loop verification and validation of UR16e based manipulator control for precise assembly and deployment tasks under uncertain terrain and visual conditions using Aruco Markers, and OpenCV.

International Lunar University Project (SSP24) **Jun 2024 – Aug 2024**

Collaboration with Dr. Jacob Cohen, Chief Scientist, NASA Ames

- Collaborated on the *International Lunar University* initiative focusing on interdisciplinary lunar research and public engagement. The project was later presented at the **International Astronautical Congress (IAC) 2024** as a non–peer-reviewed publication (DOI: 10.52202/078357-0031).

LIDAR Based Object Mapping and Identification Rover **Mar 2022 – Sept 2022**

Python, CNN, ROS, SLAM, Raspberry Pi, Keras

- An autonomous rover for mapping and object identification, for tracking space debris and warehouse management etc.

Numerically Controlled Oscillator for ISRO's SEAMS Project **February 2022 – May 2022**

Verilog, Vivado, Nxys DDR FPGA Board

- Developed a numerically controlled oscillator for spectrum analyzer calibration on the SEAMS payload.

8-bit Microcontroller using Verilog **October 2021 – March 2022**

Verilog, Nxys DDR FPGA Board, Vivado HLS

- Developed an 8-bit microcontroller for small automation tasks, using Verilog on an Nxys DDR FPGA board.

Electric Vehicle Charging Management System **Jan 2021 – Dec 2021**

Python, Django, AWS, MySQL, Apache

- Developed a cloud platform to manage and control a cluster of electric vehicle charging stations, in collaboration with IIT Delhi's CART and EVI Technologies.

Health Kiosk System for Monitoring Employees' Health **May 2020 – Dec 2020**

Raspberry Pi, Python, Flask, Sensors

- Developed a health kiosk machine deployed during COVID-19 to monitor government employees' health status in Meghalaya, India.

SEBAI: Smart Eye Based on AI **October 2019 – March 2020**

ESP32Cam, Keras, TensorFlowLite, Python, PCB Designing

- Developed an AI product for blind, deaf, and mute individuals, featuring speech-based navigation, sign language conversion, and object identification. Recognized as a finalist in the National Innovation Contest 2020.

Skills & Proficiency

Languages: Python, C, C++, Java, JavaScript, MATLAB, Verilog, VHDL, Assembly, SQL, Bash

Frameworks and Libraries: PyTorch, TensorFlow, Keras, Stable-Baselines3, OpenCV, ROS, ROS2, Flask, Django, ReactJS, MoveIt, Mujoco, Isaac Sim, Isaac Lab, OpenAI Gym, Scikit-learn

Tools and Technologies: AWS, Docker, Git, Linux, NVIDIA Jetson SDK, Vivado HLS, Cadence Design Tools, HFSS, Apache

Hardware and Embedded Systems: Jetson Orin NX, RealSense D435i, OptiTrack, FPGAs, Modular Robotic Platforms, LIDARs, Microcontrollers, Raspberry Pi, Latte Panda, PCB Designing, Sensors, IoT Devices

Research & Simulation: Reinforcement Learning, Genetic Algorithms, Sim-to-Real Transfer, Digital Twin Simulation, Multi-Agent Systems, Decentralized Control, Computer Vision, Motion Planning

Miscellaneous: Space Robotics, Modular Robotics, Autonomous Systems, Agentic Planning, Machine Learning, Neural Networks, Signal Processing, Embedded Systems Design, Cloud Computing, IoT, Cryptography, Image Processing

Awards & Honors

IAF Emerging Space Leader **2025**
International Astronautical Federation (IAF) *Sydney, Australia*

- Recognised by IAF as an Emerging Space Leader (2025); grant recipient for IAC 2025.

IAC 2025 Next Generation Plenary - Panelist **2025**
International Astronautical Federation (IAF) *Sydney, Australia*

- Invited panelist for the IAF Next Generation Plenary at IAC 2025.

Best Research Paper Award **Dec 2025**
iSpaRo 2025 (2nd International Conference on Space Robotics), Sendai, Japan

- Awarded for the paper: *"Multi-Modal Decentralized Reinforcement Learning for Modular Reconfigurable Lunar Robots."*

Japanese Government MEXT Research Scholar **2023 – Present**
Government of Japan

- Selected as a prestigious MEXT Research Scholar through Embassy Recommendation.

I-CON Award **2024**
Indian Society of Remote Sensing (ISRS), ISRO

- Awarded for an AI-enabled satellite-based air pollution monitoring and alert system, developed in collaboration with NRSC–ISRO scientists during the Space Studies Program 2024 (SSP24).

Tohoku University Dispatch Program Grant **2024**
Tohoku University

- Awarded Grant of EUR 20,000, for attending Space Studies Program 2024 by International Space University.

Innovation Ambassador **2020 – Present**
Ministry of Education, Government of India

- Selected as one of the Innovation Ambassadors at the Ministry of Education's Innovation Cell.
- Worked on enhancing mentorship skills and contributing to innovation and entrepreneurship awareness.

Finalist, National Innovation Contest 2020 **2020**
Ministry of Education, Government of India

- Secured a position among the top 10 finalists out of 30,000+ submissions in the NIC 2020 contest.

2nd Runner Up, National Level IoT Challenge 2019 **2019**
Indian Institute of Technology, Bombay

- Developed and demonstrated an IoT-based AgriTech solution for autonomous agricultural monitoring.

Research Publications

MoonBot: Modular and On-demand Reconfigurable Robot towards Moon Base Construction **2025**

IEEE Transactions on Field Robotics (T-FR)

- **DOI**: 10.1109/TFR.2025.3624346
- **Equal Authorship**
- **Authors**: Kentaro Uno*, Elian Neppel*, Gustavo H. Diaz*, **Ashutosh Mishra***, Shamistan Karimov*, A. Sejal Jain, Ayesha Habib, Pascal Pama, Hazal Gozbasi, Shreya Santra, Kazuya Yoshida

Enhancing Autonomous Manipulator Control with Human-in-loop for Uncertain Assembly Environments **2025**

IEEE CASE 2025

- **Authors**: **Ashutosh Mishra**, Shreya Santra, Hazal Gozbasi, Kentaro Uno, Kazuya Yoshida
- DOI: 10.1109/CASE.2025.11163924

Multi-Modal Decentralized Reinforcement Learning for Modular Reconfigurable Lunar Robots **2025**

IEEE iSpaRo 2025, [Best Research Paper Award]

- **Authors**: **Ashutosh Mishra**, Shreya Santra, Elian Neppel, Edoardo M. Rossi Lombardi, Shamistan Karimov, Kentaro Uno, and Kazuya Yoshida
- DOI: 10.48550/arXiv.2510.20347

Robust and Modular Multi-Limb Synchronization in Motion Stack for Space Robots with Trajectory Clamping via Hypersphere **2025**

IEEE/RSJ IROS 2025

- **Authors**:Elian Neppel, **Ashutosh Mishra**, Shamistan Karimov, Kentaro Uno, Shreya Santra, Kazuya Yoshida
- DOI: 10.1109/IROS60139.2025.11246735

Designing for Distributed Heterogeneous Modularity: On Software Architecture and Deployment of MoonBots **2025**

IEEE iSpaRo 2025

- **Authors**:Elian Neppel, Shamistan Karimov, **Ashutosh Mishra**, Gustavo Hernan Diaz Huenupan, Hazal Gozbasi, Kentaro Uno, Shreya Santra, Kazuya Yoshida
- DOI: 10.48550/arXiv.2511.01437

Design and Development of a Modular Bucket Drum Excavator for Lunar ISRU **2025**

IEEE iSpaRo 2025

- **Authors**: Simon Giel, James Michael Hurrell, Shreya Santra, **Ashutosh Mishra**, Kentaro Uno, Kazuya Yoshida
- DOI: 10.48550/arXiv.2511.00492

MLP-RF Based Hybrid ML-NN Model for GaN HEMT Parameter Estimation **2022**

International Journal of RF and Microwave Computer-Aided Engineering

- **Authors**: **Ashutosh Mishra**, Samriddhi Raut, Khushwant Sehra, Raghvendra Pratap Singh, Shweta Wadhera, Poonam Kasturi, Geetika Jain Saxena, Manoj Saxena
- DOI: 10.1002/mmce.23191

Robust and Secure Digital Image Watermarking Technique Using Arnold Transform and Memristive Chaotic Oscillators **2021**

IEEE Access Journal

- **Authors**: Khushwant Sehra, Samriddhi Raut, **Ashutosh Mishra**, Poonam Kasturi, Shweta Wadhera, Geetika Jain Saxena, Manoj Saxena
- DOI: 10.1109/access.2021.3079319

Hybrid NN Model to Predict the Radiation Damage in GaN HEMTs **2021**

IBM IEEE CAS/EDS – AI Compute Symposium

- **Authors**: Samriddhi Raut, **Ashutosh Mishra**, Khushwant Sehra, Raghvendra Pratap Singh, Shweta Wadhera, Poonam Kasturi, Geetika Jain Saxena, Manoj Saxena
- https://www.zurich.ibm.com/pdf/aics/3.16_Mishra_Ashutosh_DDUC_AICS2021_Poster.pdf
- https://www.zurich.ibm.com/thinklab/AIcomputesymposium.html

Memristor Based Cryptographic Information Processing for Secured Communication Systems **2020**

International Conference on Devices, Circuits and Systems (ICDCS)

- DOI: 10.1109/icdcs48716.2020.243573

Distributed Multi Robot Lunar Cargo Transportation via Phase Decomposed Reinforcement Learning **2026**

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

- Under Review

RoboGene: Modular DRL Framework for Reconfigurable Lunar Robots **2026**

Submission in Process: IEEE Access

Moonbot APEX: Agentic Planning and Execution for Modular Lunar Robots **2026**

Submission in Process: IEEE RA-L

Demo Link: https://youtu.be/wrkr4xLJ7zM